

$$L = \int_0^1 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$y = \left(\sqrt{\frac{2}{3}} - x^{\frac{2}{3}}\right)^{\frac{3}{2}}$$

$$\frac{dy}{dx} = \frac{3}{2} \sqrt{\frac{2}{3} - x^{\frac{2}{3}}} \cdot \left(-\frac{2}{3} x^{-\frac{1}{3}}\right) = -\frac{\sqrt{\frac{2}{3} - x^{\frac{2}{3}}}}{\sqrt{x}}$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{\frac{2}{3} - x^{\frac{2}{3}}}{x^{\frac{2}{3}}} \quad \frac{1}{2}$$

$$\sqrt{1 + \frac{\frac{2}{3} - x^{\frac{2}{3}}}{x^{\frac{2}{3}}}} = \sqrt{\frac{x^{\frac{2}{3}} + \frac{2}{3} - x^{\frac{2}{3}}}{x^{\frac{2}{3}}}} = \frac{\sqrt{\frac{2}{3}}}{\sqrt{x}}$$

$$\int \frac{\sqrt{\frac{2}{3}}}{\sqrt{x}} dx = \sqrt{\frac{2}{3}} \int x^{-\frac{1}{2}} dx = \frac{2}{2} x^{\frac{1}{2}} \sqrt{\frac{2}{3}} = \frac{2}{3} \sqrt{\frac{2}{3}} x^{\frac{3}{2}} \Big|_{\frac{\sqrt{2}}{4}}^{\sqrt{2}} = \frac{2}{3} \sqrt{\frac{2}{3}} \cdot \sqrt{2} - \frac{2}{3} \sqrt{\frac{2}{3}} \cdot \left(\frac{\sqrt{2}}{4}\right)^{\frac{3}{2}}$$

$$\frac{2 \cdot \sqrt{\frac{2}{3}} \cdot \sqrt{2}}{3} - \frac{2 \cdot \sqrt{\frac{2}{3}} \cdot \sqrt{2}^{\frac{3}{2}} \cdot \frac{1}{4}^{\frac{3}{2}}}{3} \quad \frac{2}{3} = \sqrt{\frac{2}{3}} = \sqrt{2} = 2\sqrt{2} \quad ?$$

$$\frac{21}{2} - \frac{21 \cdot \sqrt{2}}{2 \cdot 2 \cdot \sqrt{2}} = \frac{21}{2} - \frac{21}{4} = \frac{21}{4}$$

$$\frac{1}{1} - \frac{1}{2} = \frac{1}{2}$$

$$\int u dv = uv - \int v \frac{du}{dx} dx$$

$$u = x^2, dv = \cos\left(\frac{1}{2}x\right) dx \quad \frac{du}{dx} = 2x, v = 2 \sin\left(\frac{1}{2}x\right)$$

$$2x^2 \sin\left(\frac{1}{2}x\right) - \int \sin\left(\frac{1}{2}x\right) \cdot 2x dx$$

$$2x^2 \sin\left(\frac{1}{2}x\right) - (8x \cos\left(\frac{1}{2}x\right) - \int \cos\left(\frac{1}{2}x\right) 8 dx)$$

$$2x^2 \sin\left(\frac{1}{2}x\right) + 8x \cos\left(\frac{1}{2}x\right) - 16 \sin\left(\frac{1}{2}x\right) + C$$

$$u = 27 \cos(3x), v = -\frac{1}{3} e^{-3x}$$

$$\int u dv = uv - \int v \frac{du}{dx} dx$$

$$-9e^{-3x} \cos(3x) - \int 27e^{-3x} \sin(3x) dx$$

$$-\frac{27}{3} e^{-3x} \cos(3x) - \left(-\frac{1}{3}\right) \int e^{-3x} \cdot 81 \sin(3x) dx$$

$$-9e^{-3x} \cos(3x) - \left(-9e^{-3x} \sin(3x) - \int -9e^{-3x} 3 \cos(3x) dx\right)$$

$$-9e^{-3x} \cos(3x) + 9e^{-3x} \sin(3x) + (-27e^{-3x} \cos(3x))$$

$$2 \int 27 e^{-3x} \cos(3x) dx = 9e^{-3x} (\sin(3x) - \cos(3x))$$

$$= \frac{9}{2} e^{-3x} (\sin(3x) - \cos(3x))$$

$$\lim_{b \rightarrow -\infty} \int_b^0 e^{-|3x|} dx = \lim_{b \rightarrow -\infty} \left. -\frac{1}{13} e^{-13x} \right|_b^0 \quad \text{for } -\infty \leq x \leq 0$$

$$= \frac{1}{13} e^{-13x} - \frac{1}{13} e^{-13 \cdot \infty}$$

$$= \frac{1}{13} \cdot 1 - 0$$

$$\lim_{b \rightarrow -\infty} \int_b^0 x e^x dx \quad \begin{matrix} u = x \\ v = e^x \end{matrix} \quad uv - \int v \frac{du}{dx} x$$

$$= 1 - x e^x - \int_b^0 e^x dx$$

$$= 1 - x e^x \Big|_b^0 - e^x \Big|_b^0$$

$$= 1 - 0 - b e^b - (1 - e^b)$$

$$\lim_{b \rightarrow -\infty} = 0 - \infty \cdot 0 - 1 - 0 = -1$$

$$\lim_{c \rightarrow 100^-} \int_0^c \frac{1}{\sqrt{100-x}} dx = -2 \sqrt{100-x} \Big|_0^c$$

$$= -2 \sqrt{100-c} + 2 \sqrt{100-0}$$

$$= 0 + 2 \sqrt{100} = 20$$